

WQU MSFE 690 CAPSTONE PROJECT – MODULE 4 DESIGN

Project Proposal – Student Group 17830

Group Members: Ting Ting Han, Henry Mayambu, Jonathan Matura

Research Title: ESG Disclosure and Volatility in JSE Mining: Risk-Adjusted Performance 2019-2023

Repo: esg-jse-mining-capstone | Main File: `esg\_jse\_mining\_analysis.ipynb`

1. Research Question

Do higher ESG disclosure scores reduce volatility and improve risk-adjusted returns for JSE mining companies (2019-2023)?

H1: Higher ESG → lower conditional volatility (GARCH).

H2: Higher ESG → lower Conditional Value-at-Risk (CVaR 95%).

H3: Causal effect: ESG → Volatility after controlling for market capitalization and commodity exposure.

2. Motivation

Mining is core to the South African economy, contributing ~ 8% of GDP, yet faces intense ESG scrutiny. In June 2022, the JSE issued ESG Guidance Letter requiring sustainability disclosure. There is no consensus whether ESG disclosure reduces financial risk in emerging market mining, where commodity cycles dominate. This project quantifies ESG alpha for risk management.

3. Data

`JSE\_sample\_data (2019-2023) - 12 firms, 60 months` from JSE. Variables: Ticker, Date, Close, Return, ESG Score, MktCap. File: `JSE\_sample\_data (1).csv`. We construct monthly log returns, ESG quartile dummies, and market cap control. Data cleaned in `esg\_jse\_mining\_analysis.ipynb`.

4. Methodology

a) **Panel Fixed Effects:** $\text{Return}_{it} = \alpha_i + \beta_1 \text{ESG}_{it} + \beta_2 \ln(\text{MktCap})_{it} + \varepsilon_{it}$ – to test risk-adjusted return.

b) **GARCH(1,1):** $\sigma^2_t = \omega + \alpha_1 \varepsilon^2_{t-1} + \beta_1 \sigma^2_{t-1}$. Estimated separately for High-ESG vs Low-ESG quartile to test H1. Graphs will have X-axis: Date (2019-2023), Y-axis: Conditional Volatility (%), Title: GARCH Volatility by ESG Quartile.

c) **CVaR 95%:** $\text{CVaR} = E[\text{Loss} \mid \text{Loss} > \text{VaR}_{95\%}]$. Compare High vs Low ESG portfolio.

d) **Causal Bayesian Network (pgmpy):** DAG: ESG Score → Return / Volatility ← MktCap. Test causal strength for H3.

e) **Long High-ESG / Short Low-ESG backtest:** Equal-weighted, monthly rebalance, transaction cost 10bps. Performance metrics: Sharpe, Max Drawdown.

5. Pain-Points in Current Financial Engineering Strategies

Current financial engineering approaches have three gaps: (1) Volatility models for mining stocks attribute volatility solely to commodity price and FX, ignoring ESG disclosure as a volatility dampener. (2) ESG integration uses simple correlation, not conditional heteroskedasticity (GARCH) or tail risk (CVaR), leading to underestimation of ESG benefits in left-tail events. (3) Greenwashing risk: without causal inference, funds cannot distinguish if high-ESG firms are less risky or simply larger firms. Our project addresses this by combining GARCH + CVaR + Causal Network.

6. Possible Obstacles and Impediments

(1) Small sample: 12 firms x 60 months = 720 obs limits power. (2) ESG scores self-reported, inconsistent before JSE 2022 guidance, scoring methodology changes. (3) Confounders: Coal vs Gold vs Platinum exposure, load-shedding events, commodity super-cycle 2021-2022. (4) Survivorship bias: delisted mining firms not in sample. (5) pgmpy causal assumptions require no hidden confounders – we mitigate with sensitivity analysis. (6) JSE illiquidity may bias Close prices.

7. Other Possible Real-Life Applications

Beyond JSE mining: (1) Zambian Copperbelt – framework transferable to KCM, Mopani, First Quantum for ESG-linked loan pricing. (2) Institutional asset management – NAPSA and other African pension funds for ESG-tilted mining portfolios. (3) Insurance underwriting – tail-risk estimates for mine

rehabilitation liability. (4) Corporate treasury – hedging cost reduction for high-ESG miners. (5) Regulatory – evidence for JSE to mandate assurance of ESG disclosure.

8. Incorporation of Feedback from Previous Assignments

Per feedback received in M3 Proposal Review: (1) We narrowed scope from all JSE to mining sector only and period to 2019-2023 to align with JSE ESG guidance. (2) We added CVaR 95% High vs Low ESG comparison as requested to capture tail risk. (3) We added explicit causal inference via Bayesian Network instead of only correlation. (4) We specified repository structure with requirements.txt and CONTRIBUTORS.md.

9. Expected Contribution

Quantify ESG alpha for JSE mining, provide open-source risk management framework combining panel, GARCH, CVaR and causal inference, and publish reproducible open-source repo with documentation for emerging market ESG research.

10. Division of Labor

Ting Ting Han: Literature review, data validation, Panel Fixed Effects.

Henry Mayambu: GARCH(1,1) / CVaR estimation, GitHub Repo management, MLA bibliography.

Jonathan Matura: Causal Bayesian Network (pgmpy) & Long/Short backtest, README.

All: Joint writing and QA.

11. Repo Contents & Preliminary Results

README.md, requirements.txt, data: `JSE\_sample\_data (1).csv`, `esg\_jse\_mining\_analysis.ipynb` (updated after M3 with preliminary GARCH plot showing High-ESG quartile mean volatility 22.3% vs Low-ESG 31.7%, initial CVaR 95% High-ESG -8.2% vs Low-ESG -12.5%), /png folder with labeled figures, CONTRIBUTORS.md. Code shows preliminary results and initial inferences for 20-point Goals criteria.

12. References – MLA Format

Johannesburg Stock Exchange. "JSE Sustainability and Climate Disclosure Guidance." JSE, June 2022.

Bollerslev, Tim. "Generalized Autoregressive Conditional Heteroskedasticity." Journal of Econometrics, vol. 31, 1986, pp. 307-327.

Rockafellar, R. Tyrrell, and Stanislav Uryasev. "Conditional Value-at-Risk for General Loss Distributions." Journal of Banking & Finance, vol. 26, 2002.